

Vertical Antenna Tilt Optimization for LTE Base Stations

Harald Eckhardt, Siegfried Klein, and Markus Gruber

Alcatel-Lucent Bell Labs
Lorenzstr. 10, 70435 Stuttgart, Germany
Harald.Eckhardt@Alcatel-Lucent.com

Abstract—An optimal vertical antenna tilt in a wireless access network plays a key role for coverage and capacity in the system. The autonomous (re-)optimization of tilts bears a large saving potential for operators as manual intervention is particularly costly. In this paper, we present a heuristic variant of the gradient ascent method to continuously optimize antenna tilts with quick convergence. We show that the average spectral efficiency in the system increases by 10% and that the spectral efficiency at the cell edge (5% quantile) increases by 100% after optimization. Changing conditions during the operational phase of the network, as for instance cell outage, can successfully be detected and compensated by an autonomous re-optimization.

Index Terms—self-optimization, cell outage compensation, tilt optimization, coverage and capacity optimization, gradient ascent

I. INTRODUCTION

The self-optimization of wireless access network elements is an essential functionality that substantially saves cost for the network operator. In this context, the optimization of the vertical antenna tilt (downtilt with respect to the horizontal plane) of a base station is particularly delicate as manual intervention is costly. The self-optimization of antenna tilts has been addressed in a number of publications. Advantages of automatic downtilting were already discussed for 3rd generation cellular networks [1][2]. [3][4][5] elaborated on tilt optimization in connection with power adaptations, whereas [6] mentioned the antenna tilt as a important parameter for energy efficiency. [7] treated tilt optimization in the context of a multi-objective problem involving further parameters. This aspect was further investigated by [8][9] with a special emphasis on the interplay between different optimization processes. Eventually, [10] provided a general mathematical framework for self-optimizing functionalities including the antenna tilt. In this paper we propose a distributed heuristic variant of the gradient ascent method for the tilt optimization of a system in operational mode that we evaluate by means of an LTE-like simulation environment.

II. HEURISTIC VARIANT OF GRADIENT ASCENT METHOD

In this section, we will present a heuristic variant of the gradient ascent method in order to optimize the vertical antenna tilt of an OFDMA base station with respect to a given utility metric.

A. The Metric to be Optimized

Let us assume base stations with three sectors each and an overall system topology consisting of a set P of sectors with a total of $|P|$ antennas for $|P|$ sectors. In our analysis, this set is defined as $P = \{3, 4, \dots, 59\}$. The utility metric we refer to is defined as

$$U = \frac{1}{|P|} \sum_{m \in P} (w_{avg} s_{m,avg} + w_{edge} s_{m,edge}) \quad (1)$$

where $s_{m,avg}$ is the average spectral efficiency over the entire sector m and $s_{m,edge}$ is the 5% quantile of the distribution of the spectral efficiency over the entire sector m . The spectral efficiency can be derived from the SINR by applying the LTE coding schemes [11] and by measuring over a sufficiently long interval until the optimization area is adequately covered. Coverage holes breaking the signaling link shall be added to the statistics after link re-establishment, which can occur in another sector.

$w_{avg} = \frac{1}{\text{bits/s/Hz}}$ and $w_{edge} = \frac{10}{\text{bits/s/Hz}}$ are weights (tunable by the operator) that give sector edge users, which have low spectral efficiencies in the first place, a reasonable importance. The utility metric is dimensionless.

B. The Optimization Algorithm

In the following we will present an algorithm that uses real measurements from terminals as input for optimization. This approach does not require a specific model nor data on how the topology of the network was planned as the operational system is used to test the new tilt settings.

Each antenna $m \in P$ can adopt an individual tilt so that the tilts of all antennas can be described as a tilt vector

$$\vec{\phi} = \begin{pmatrix} \vdots \\ \phi_m \\ \vdots \end{pmatrix} \text{ with } |P| \text{ dimensions. At the beginning of the}$$

optimization, all angles are initialized to 11°, which would be the optimum if all antennas had equal tilt angles.

For a given instant t , an arbitrary sector $j_t \in P$ can be chosen as starting point which then acts as the *center sector* of the cluster $C_j = \{j_t\} \cup \{m | m = \text{neighbor}(j_t)\}$ where neighbors are defined to be those sectors that have the same

geometric relation to the center sector j_t as the colored sectors have to the red sector in Fig. 1. More intuitively, the definition of neighbors is based on the mutual interference of the sectors in the network. The cluster $C_j \subseteq P$ is thus a subset of the overall system P . The tilt vector at instant t is denoted φ_t

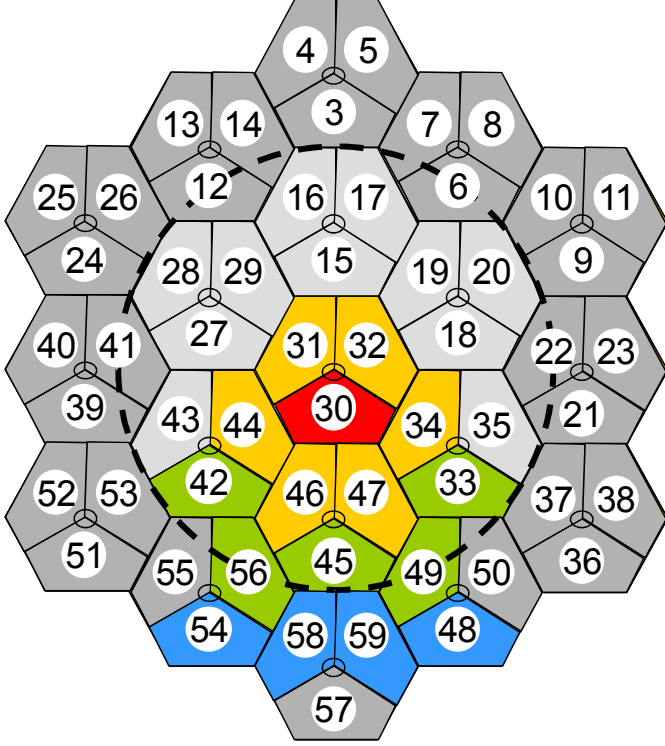


Fig. 1: Schematic view on the system environment. From an arbitrary center sector j_t (in this case sector 30 in red) the cluster C_j is defined to consist of all colored sectors. This geometrical principle is applied to all potential center sectors by shifting and rotating this pattern; for this purpose a “wrap around” according to [13] at the borders is assumed.

Within C_j , we consider the gradient of the utility U with respect to the individual tilts φ_i with the individual differentials

$$\frac{\partial U}{\partial \varphi_i} = \begin{cases} \frac{U(\varphi_i + \Delta_i) - U(\varphi_i - \Delta_i)}{(\varphi_i + \Delta_i) - (\varphi_i - \Delta_i)} & \text{for } i \in C_j \setminus S \\ 0 & \text{else} \end{cases}$$

where

$$\Delta_i = \begin{bmatrix} \vdots \\ \Delta \delta_{im} \\ \vdots \end{bmatrix} \text{ for } m \in P \text{ and the Kronecker } \delta_{im}$$

and

$$S = \{i \mid U(\varphi_i) < U(\varphi_i + \Delta_i) \text{ and } U(\varphi_i) < U(\varphi_i - \Delta_i)\}$$

which is the set of sectors where a change of the angle by Δ_i in either direction indicates a minimum. Then the maximal absolute value of all tilt differentials

$$c = \max_{i \in C_j} \left| \frac{\partial U}{\partial \varphi_i} \right|$$

is used as normalization factor when determining the candidate vertical antenna angles:

$$\varphi_{m,k} = \begin{cases} \varphi_m + \frac{k}{c} \frac{\partial U}{\partial \varphi_m} + 0.5 & \text{for } m \in C_j \text{ and } k \in K \\ \varphi_m & \text{for } m \in P \setminus C_j \text{ and } k \in K \end{cases}$$

where $K = \{1, 2, \dots, k_{\max}\}$. The original set K is further modified to a set K^* containing additional real number elements that are intermediate between the ones yielding the best and second best utilities.* Thus, we obtain $|K^*|$ new tilt vectors, indexed by k :

$$\bar{\varphi}_k = \begin{bmatrix} \vdots \\ \varphi_{m,k} \\ \vdots \end{bmatrix} \text{ for all } m \in P \text{ and } k \in K^*$$

and thanks to the normalization factor c the largest change of an individual angle is limited to $\pm k_{\max}$ degrees.

Eventually, the best-performing tilt vector out of the tested angles φ : $\varphi_t, \varphi_t + \Delta_i, \varphi_t - \Delta_i, \varphi_k$

is determined and taken as new tilt vector

$$\bar{\varphi}_{t+1} = \arg \max_{\varphi} (U(\varphi_i), \{U(\varphi_i + \Delta_i) \mid i \in C_j\}, \{U(\varphi_i - \Delta_i) \mid i \in C_j\}, \{U(\varphi_k) \mid k \in K^*\}) \quad (2)$$

Further, a new center sector j_{t+1} is determined by a simple algorithm giving all sectors an equal chance of being center sector and letting the clusters cover the entire area in the first few steps already: $j_{t+1} = \text{mod}(j_t - P_{(1)} + p_0, |P|) + P_{(1)}$

Here we assume that the elements of P are consecutively enumerated starting with $P_{(1)} = 3$ and that $p_0 = 13$.

Finally, the entire procedure described above is iteratively resumed with the new starting points $\bar{\varphi}_{t+1}$ and j_{t+1} (and thus a new cluster). Equation 2 guarantees a monotonous increase of $U(\varphi_i)$, because it contains the (up till now) best $U(\varphi_i)$. The entire procedure is repeated for all clusters until the point where no further optimization can be reached during a full sequence $|P|$ of clusters. If the ambient system conditions change, the procedure will continue to test tilts and track the optimum.

III. SYSTEM SIMULATOR

The proposed algorithm was evaluated by means of an LTE system simulator [12]. In the simulation environment, a three-sector site is surrounded by two tiers of neighboring sites such that there is a total of 19 sites or 57 sectors, correspondingly (Fig. 1), with a “wrap around” [13] implemented. The electrical tilt 3D-antenna model corresponds to the simulation assumptions in [12] and does not only have a vertical, but also a horizontal pattern. Simulation details can be found in Table 1. Further concrete values related to the presented algorithm

* For a further optimization it is conceivable to also dynamically remove some elements of K that are less likely to provide a high utility value.

are $\Delta_i = 2^\circ \forall i \in C_j$ and $K = \{1, \dots, 6\}$. $k_{\max}$ is chosen to be 6 as it was observed that a higher $k_{\max}$ does not give any benefits in equation 2.

TABLE I. SIMULATION ASSUMPTIONS.

Parameter	Assumption
Site Type	3-sectorized
Inter-site distance	500m with displacement
Propagation Model	$128.1 + 37.6 * \log_{10}(d)$ (Distance d in km)
Minimum distance between terminal and base station	35 m
Carrier Frequency	2 GHz
System Bandwidth	10 MHz
Thermal noise	-104 dBm at 10 MHz bandwidth
Base station transceiver power	46 dBm
Base station antenna horizontal pattern	70° at 3 dB beam width, 25 dB backward attenuation
Base station antenna vertical pattern	10° at 3 dB beam width, 20 dB backward attenuation
Base station overall antenna pattern	25 dB backward attenuation
Base station antenna maximum transmission gain	15 dBi
Base station height	15 m
Terminal height	1.5 m

Base stations have an average inter site distance of 500 m, but their actual positions are displaced by a distance that is equally distributed between 0 m and 200 m in a random direction. The measurement interval is 100 h divided by the number of terminals and based on a random-walk mobility model. Spatially correlated shadow fading [14] with a decorrelation length of 50 m and a standard deviation of 8 dB is applied. The outage is realized by decreasing the base station power to -200 dBm.

IV. SIMULATION RESULTS

In this section we discuss the performance of the tilt optimization algorithm presented in Section II. In this context there are four landmark time instants, namely $t=1$ for the start situation before any optimization took place; $t=74$ after optimization; $t=75$ right after the occurrence of an outage of the base station hosting sectors 30, 31, and 32; and $t=104$ after re-optimization following the outage. We considered the occurrence of an outage problem in order to show that the algorithm behaves in a benign way and converges quickly. As a matter of fact, besides the initial rollout of the system, outage is a major reason why tilt optimization should be able to act autonomously and continuously.

Fig. 2 illustrates the behaviour of the overall utility over time instants. In particular, the drop of the utility at the occurrence of the outage can be observed at $t=75$. Eventually, the utility recovers, but at a lower value than before since the outage has reduced the available radio resources in the system.

Fig. 3 and Fig. 5 show best server plots and SINR plots for these four time instants. The best server plots indicate what sector is dominant in terms of SINR, whereas the SINR plot reveals coverage issues in the system. A magnification of two sample areas in the system is shown for each time instant to illustrate the changes in terms of SINR that are achieved: for all time instants, the left area is the one between sectors 18 and

22 that has some coverage problems at the sector edge and that becomes well optimized; the right area is the outage area of sectors 30 to 32 that is served by the surrounding sectors and is comparable to other sector edges after optimization.

Fig. 5 shows that the outage is well compensated. Note that generally the optimization algorithm does not have any location information - the high spatial resolution in the figures is only to illustrate the effects of the optimization. Black pixels in the SINR plots indicate that no measurement data is available for these locations as the mobility model placed no terminals there during the measurement interval. Coverage holes are indicated by red SINR values.

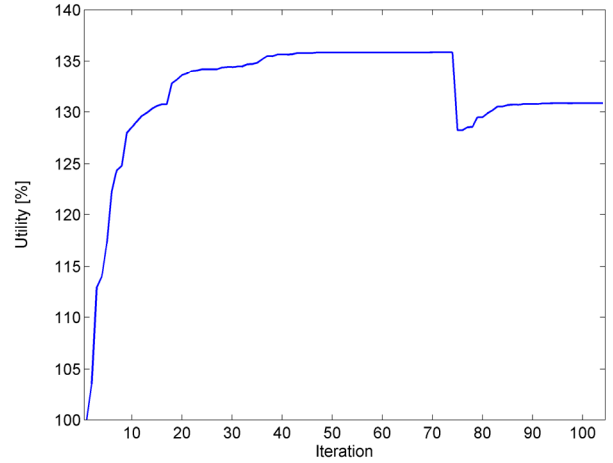


Fig. 2: Change of overall utility over time instants. Outage occurs at $t=75$.

Fig. 4 shows how the vertical antenna tilt is distributed over the sectors after optimization. The antennas adopt tilt angles ranging from 4° to 19° , whereas in most cases there are only small changes with respect to the initial angle of 11° .

Fig. 6 shows the distribution of the per-sector utilities at the landmark time instants. The mean utility (i.e., the overall utility as defined in equation 1 considers all 57 sectors even if some of them are affected by an outage. This in turn means that the utility after an outage is worse than before the outage and this reflects the fact that a lower number of sectors naturally comes along with a lower number of radio resources. Moreover, the mean utility is obviously lowest before any optimization and reaches the highest value after optimization. When the individual sectors are sorted in order of increasing utility, the original tilt configuration before any optimization performs clearly worse than any other configuration. The configurations at the landmark time instants $t \neq 1$ only show substantial differences for a limited number of sectors (directly or indirectly) affected by the outage.

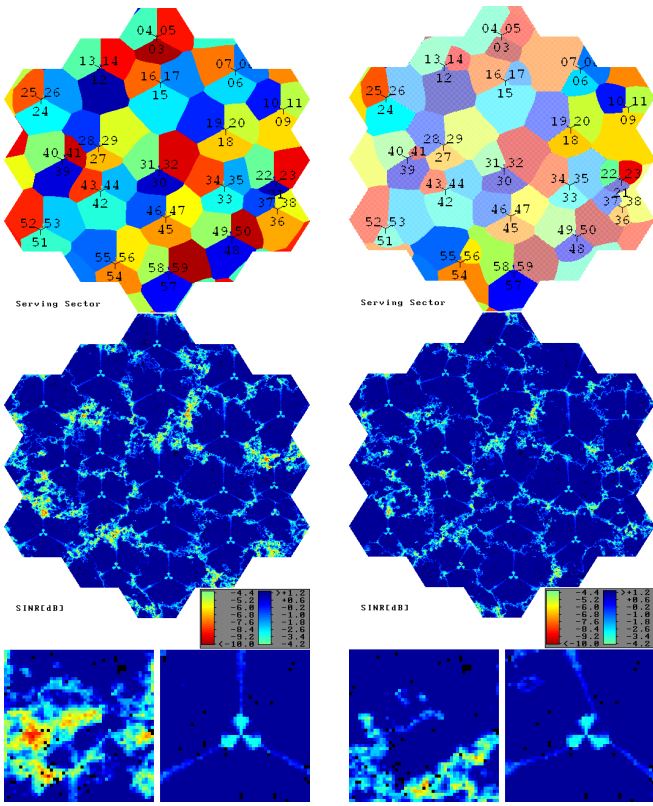


Fig. 3: Simulation environment before outage. a) Best server plot. b) SINR plot in dB (color coded according to legend) before optimization (left, $t=1$) and after optimization (right, $t=74$). c) Magnification of square area in SINR plot.

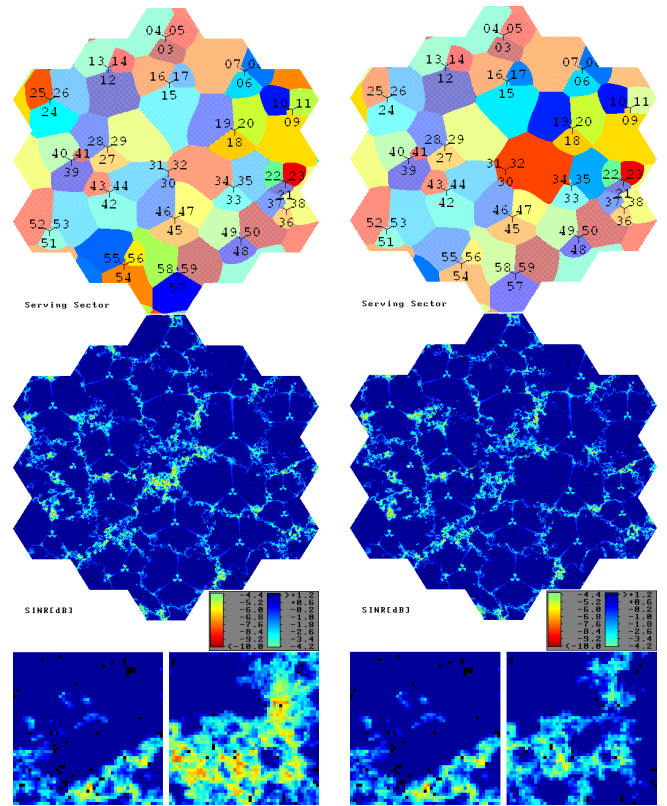


Fig. 5: Simulation environment after outage of sectors 30 to 32. a) Best server plot. b) SINR plot in dB (color coded according to legend) before re-optimization (left, $t=75$) and after re-optimization (right, $t=104$). c) Magnification of square area in SINR plot.

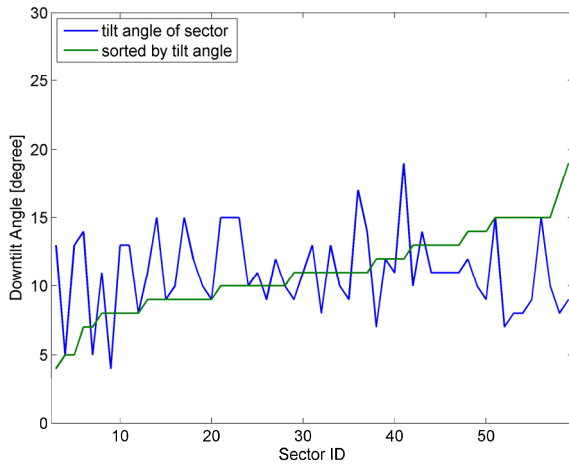


Fig. 4: Vertical antenna angles after optimization ($t=74$, i.e., before outage).

Fig. 7 shows performance gains and degradations, respectively, when applying the proposed tilt optimization algorithm (before outage). The average spectral efficiency increases by approximately 10%. 46 sectors benefit from an improvement, whereas only 11 sectors experience a degradation. The sector edge spectral efficiency improves by a solid 100%. In this case 52 sectors benefit with a mere 5 sectors with less favourable performance.

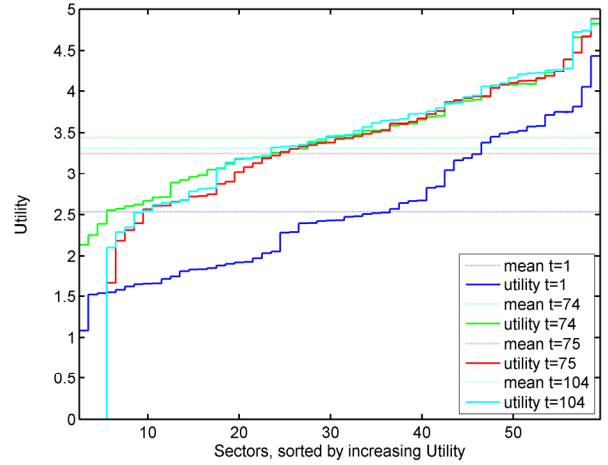


Fig. 6: Utility per sector before optimization (instant $t=1$), after optimization right before outage ($t=74$), right after outage ($t=75$), and optimized after outage ($t=104$).

Fig. 8 shows the CDF of the spectral efficiency for two sample sectors. From such distributions, the two spectral efficiencies $s_{m,avg}$ and $s_{m,edge}$ as used in equation 1 can be deduced: the average spectral efficiency (indicated by markers on the curves) and the edge spectral efficiency represented by the 5% quantile. Sector 21 belongs to a site that is relatively close to its neighbor site, so the start tilt of 11° at $t=1$ is far from the optimum angle of 15° (see Fig. 4 for the sector tilt values) and the optimization yields a large tilt difference till

$t=74$. The outage does not affect this sector, as it is too far away from the outage area. By contrast, sector 34 is one of the sectors that substantially compensates for the site affected by the outage. Initially, the optimization from $t=1$ to $t=74$ changed the tilt only slightly from 11° to 10° and achieved some improvements, but the main change (4° at $t=104$) occurs when this sector has to compensate for the outage (of its neighbor sectors 30 to 32). This is particularly delicate as the antenna has to reach an area that it is not arranged for. The minimum and maximum spectral efficiency is limited by the available code rates of the underlying LTE radio layer and the resolution is 0.1 bits/s/Hz.

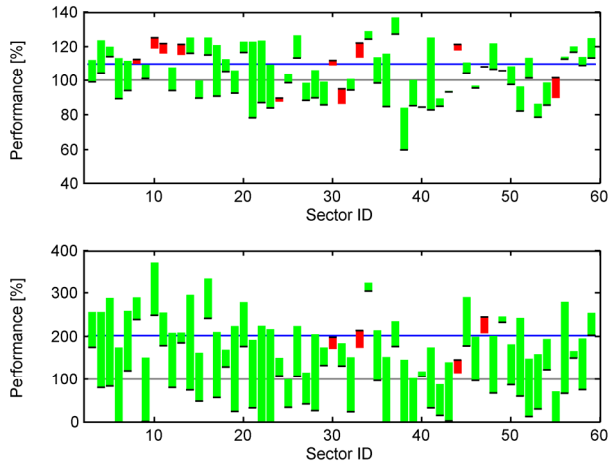


Fig. 7: Performance gain of proposed optimization algorithm (between $t=1$ and $t=74$); top: with respect to total spectral efficiency, bottom: with respect to sector edge spectral efficiency. Green bars represent improvements from the bottom to the top of the bar and red bars represent performance degradations from the top to the bottom of the bar. Blue line: average performance before optimization, grey line: average performance after optimization.

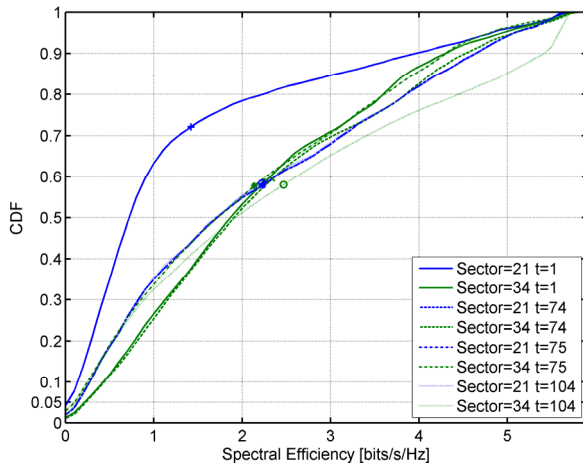


Fig. 8: Cumulative probability density functions (CDF) of the spectral efficiency for selected sectors (21 in blue and 34 in green, sector numbers as in Fig. 1) and the landmark time instants t .

V. CONCLUSION

In this paper we have proposed a heuristic variant of the gradient ascent algorithm to improve the overall and sector

edge spectral efficiency by changing the vertical antenna tilts of base stations. In an LTE-like simulation environment we showed that the average sector spectral efficiency can be improved by 10% while the sector edge spectral efficiency can even be improved by 100%. The proposed algorithm can thus be favourably employed for an automatic adjustment of antenna tilts in an operational system, even in case of cell outages, and will therefore substantially reduce network operating costs. As little communication is needed between nodes, the algorithm can be implemented in both distributed and centralized network architectures.

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